Appendix

Basics in financial mathematics, statistics, accounting

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Introduction

- 30017/30366 Corporate Finance requires some basics in mathematics and statistics. Bocconi students take Financial Mathematics, Accounting, and Financial Markets and Institutions as prerequisite courses. These cover all concepts that we need for the course.
- Having taken these courses or their equivalents is advised. This slide set covers some of the basics that you should be familiar with. These include
  - from financial mathematics the time value of money, annuities and perpetuities;
  - basics in statistics such as probabilities, variances/covariances and ordinary least squares;
  - basics in accounting such as being able to read information contained in balance sheets, income statements, cash flow statements.
- These slides have been prepared by Professors Viktar Fedaseyeu and Hannes Wagner

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Outline

1 Statistics: review of some concepts

2 Accounting: review of some concepts

3 Time value of money and present values

4 Compounding

5 Bond yields, prices, and duration

6 Risk-return trade-off

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Draw from online resources

- In addition to the basics covered here, you should draw from freely available online resources whenever you feel you are lacking knowledge of basic concepts.
- MOOCs and similar formats can teach the basics just as well or better than we could cover them in class.
- I recommend the courses available on Coursera and, especially, those available on KhanAcademy:
  - Math – Probability and statistics
  - Economics and finance – Finance and capital markets

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Why do we care?

- Financial markets produce large amounts of data. Understanding those data is necessary to make money. It is that simple.
- Large amounts of data are often compressed into more easily understandable summaries. These give a sense of the content. There are many ways to present data, and they can lead to different impressions.
- We will review some of the most widely used summary statistics in finance.
- This review is not meant to replace a base-level introduction into statistics and econometrics, it is simply a refresher for concepts we will use frequently.

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Returns and expected returns

- Consider a random return, such as that generated by a firm's stock over time, we call it $\tilde{r}$.
- When we talk about the return of a financial instrument, such as stock, what we really mean is the *holding-period return* over a specific period during which we hold (own) the instrument.
- The holding period return, for a dividend-paying stock for example, would then be

$$r = \frac{FinalPrice - StartingPrice + Dividend}{StartingPrice}$$

- The **mean** return, $\bar{r}$, is the expected value of those returns, i.e. the probability-weighted average of the rates of return over time

$$\bar{r} = E(\tilde{r}) = \sum_t p(t) r(t)$$

- where t are the possible states of the world, p(t) is the probability of each state, and r(t) is the return in that state.

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Variance

- The **variance** of a return is the expected value of the squared demeaned return outcomes, i.e.

$$\mathrm{var}(\tilde{r}) = E[(\tilde{r} - \bar{r})^2]$$

Variance is usually denoted as σ^2 or $\sigma^2(\tilde{r})$

- Example
  - Compute the variance of the return of a $400,000 investment in SINTEL stock. Assume that over the next period SINTEL earns 20% 8/10 of the time, loses 10% 1/10 of the time and loses 40% 1/10 of the time.
  - The mean return is 11% (=.8*.2+.1*(-.1)+.1(-.4))
  - To obtain variance, the demeaned returns are 9%, -21% and -51%. Therefore variance is

$$\sigma^2 = 0.8(0.09)2 + 0.1(-0.21)2 + 0.1(-0.51)2 = 0.0369$$

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A note on calculating variance

- When variance (or std dev) is estimated from a sample of observed returns, the squared deviations are usually divided by N-1, where N is the number of observations.
- Dividing by N-1 rather than by N corrects for the loss of a degree of freedom.
- The variance formula then is

$$Var(\tilde{r}_m) = \frac{1}{N-1}\sum_{t=1}^{N}(\tilde{r}_{mt} - r_m)^2$$

$$where$$

$$\tilde{r}_{mt} = \text{return in period } t$$

$$r_m = \text{mean of } \tilde{r}_{mt}$$

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Standard deviation and covariance

- **Standard deviation** is the square root of variance, denoted

$$\sigma \text{ or } \sigma(\tilde{r})$$

- It is typically reported in percent per year (if returns are reported in percent per year).
- We also have

$$\sigma(x\tilde{r}) = x\sigma(\tilde{r})$$

- **Covariance** is a measure of relatedness. The covariance between two returns is the expected product of their demeaned outcomes: $\sigma_{12} = E[(\tilde{r}_1 - \overline{r}_1)((\tilde{r}_2 - \overline{r}_2)]$

- In order to calculate σ_{12} we need the joint distribution of returns of both stocks

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Correlations

- Variance is a special case of the covariance:

$$\mathrm{cov}(\tilde{r},\tilde{r}) = \mathrm{var}(\tilde{r})$$

- The **correlation** between two returns is the covariance between the two returns divided by the product of their std deviations:

$$\rho(\tilde{r}_1,\tilde{r}_2) = \frac{\mathrm{cov}(\tilde{r}_1,\tilde{r}_2)}{\sigma_1\sigma_2}$$

- Intuition for ρ: Rescaled covariance between -1 and +1.
  - ρ=1: Perfect positive correlation, ρ=-1: Perfect negative correlation
- **Translating correlations into covariances:**

$$\mathrm{cov}(\tilde{r}_1,\tilde{r}_2) = \rho(\tilde{r}_1,\tilde{r}_2)\sigma_1\sigma_2$$

$$\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$$

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Example: Obtaining estimates from historical data

- Below is an example of historical returns for firm AAA and the S&P 500 index, i.e. their joint return distribution. Verify that you can obtain (on paper!) stddev,var,cov,corr as indicated.

	Firm AAA returns		S&P 500 Index returns
t=1	0.12		0.08
t=2	0.10		0.12
t=3	-0.05		-0.03
Variance	0.0086		0.0060
StandardDeviation	0.0929		0.0777
	Covariance		**Correlation**
	0.0067		0.9330

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Ordinary least squares regression

- A simple regression is an extension of the correlation/covariance concept. It attempts to explain one variable, which is called the dependent variable, using the other variable, called the independent variable.
- Keeping with statistical tradition, let Y be the dependent variable and X be the independent variable. If the two variables are plotted against each other on a scatter plot, with Y on the vertical axis and X on the horizontal axis, the regression attempts to fit a straight line through the points in such a way as the minimize the sum of the squared deviations of the points from the line.
- Consequently, it is called ordinary least squares (OLS) regression.

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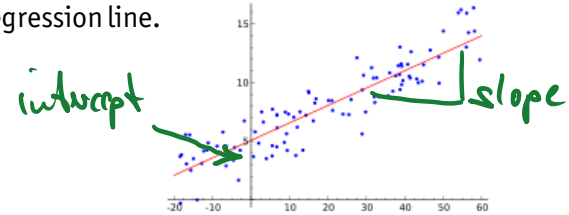
- When such a line is fit, two parameters emerge – one is the point at which the line cuts through the Y axis, called the intercept of the regression, and the other is the slope of the regression line.



- OLS regressions can be run in many applications, including Excel. There are **many** online tutorials for setting up OLS and interpreting results. **Make sure to review at least one of those if you have not previously run a regression with actual data yourself.**

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Why do we care about accounting?

- Running a large firm is a team effort, involving many parties - shareholders, lenders, directors, management, employees and so on. All have a stake in a firm's success and need to monitor its progress.
- Because of this, a firm prepares regular financial accounts and arranges for an independent auditor to certify that these accounts present a «true and fair view».
- Historically, accounting used to be less formal, more national in focus, less standardized.
- We will not discuss the details of accounting practice. But we will refer to financial statements, this is where we get some of our data, and we therefore need to understand the main features.

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BALANCE SHEET FOR PEPSICO, INC.
(Figures in millions of dollars)

Assets	1998	1997
Current assets		
Cash and equivalents	311	1,928
Marketable securities	83	955
Receivables	2,453	2,150
Inventories	1,016	732
Other current assets	499	486
Total current assets	4,362	6,251
Fixed assets		
Property, plant, and equipment	13,110	11,294
Less accumulated depreciation	5,792	5,033
Net fixed assets	7,318	6,261
Intangible assets	8,996	5,855
Other assets	1,984	1,734
Total assets	22,660	20,101

Liabilities and Shareholders' Equity	1998	1997
Current liabilities		
Debt due for repayment	3,921	0
Accounts payable	3,870	3,617
Other current liabilities	123	640
Total current liabilities	7,914	4,257
Long-term debt	4,028	4,946
Other long-term liabilities	4,317	3,962
Total liabilities	16,259	13,165
Shareholders' equity		
Common stock and other paid-in capital	1,195	1,343
Retained earnings	5,206	5,593
Total shareholders' equity	6,401	6,936
Total liabilities and shareholders' equity	22,660	20,101

Full annual report at www.pepsico.com/Annual-Reports/1998/annual1998.pdf

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The Balance Sheet

Definition

Financial statements that show the value of the firm's assets and liabilities at a particular point in time *(from an accounting perspective).*

Imagine an accountant taking a snapshot of a firm's value, as if the firm stood momentarily still.

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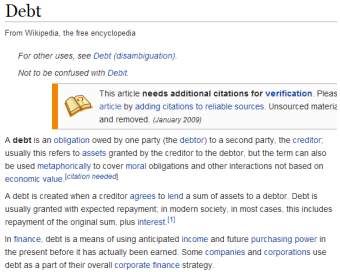
Assets vs liabilities

- The left-hand side of the balance sheet shows the assets of a firm. The right-hand side of the balance sheet shows where the money to buy the assets came from.
- What is a liability then?
  - First definition: The money owed by the company.
  - Second definition: The overall amount of financing raised by the firm, consisting of debt and equity.



Debt

From Wikipedia, the free encyclopedia

For other uses, see Debt (disambiguation).

Not to be confused with Debt.

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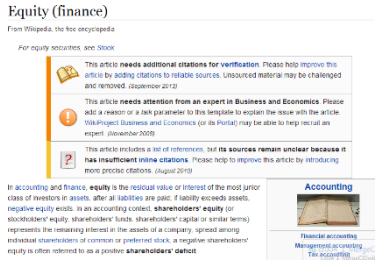
A **debt** is an obligation owed by one party (the debtor) to a second party, the creditor; usually this refers to assets granted by the creditor to the debtor, but the term can also be used metaphorically to cover moral obligations and other interactions not based on economic value.[citation needed]

A debt is created when a creditor agrees to lend a sum of assets to a debtor. Debt is usually granted with expected repayment; in modern society, in most cases, this includes repayment of the original sum, plus interest.[1]

In finance, debt is a means of using anticipated income and future purchasing power in the present before it has actually been earned. Some companies and corporations use debt as a part of their overall corporate finance strategy.



Equity (finance)

From Wikipedia, the free encyclopedia

For equity securities, see Stock.

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In accounting and finance, **equity** is the residual value or interest of the most junior class of investors in assets, after all liabilities are paid; if liability exceeds assets, negative equity exists. In an accounting context, **shareholders' equity** (or stockholders' equity, shareholders' funds, shareholders' capital or similar terms) represents the remaining interest in the assets of a company, spread among individual shareholders of common or preferred stock, a negative shareholders' equity is often referred to as a positive **shareholders' deficit**.

Accounting

Financial accounting

Management accounting

Tax accounting

The Balance Sheet

The Main Balance Sheet Items

Current Assets

Securities

Cash &

Receivables
Inventories

+

Fixed Assets

Tangible Assets
Intangible

Assets

=

Current Liabilities

Payables
Short-term Debt

+

Long-term Liabilities

+

Shareholders' Equity

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INCOME STATEMENT FOR PEPSICO, INC., 1998
(Figures in millions of dollars)

Net sales	$22,348
Cost of goods sold	9,330
Other expenses	291
Selling, general, and administrative expenses	8,912
Depreciation	1,234
Earnings before interest and taxes (EBIT)	2,581
Net interest expense	321
Taxable income	2,260
Taxes	270
Net income	1,990
Allocation of net income	
Addition to retained earnings	1,233
Dividends	757

 © Hannes Wagner

The Income Statement

Definition

Financial statement that shows the revenues, expenses, and net income of a firm over a period of time *(from an accounting perspective).*

If the balance sheet is like a snapshot, the income statement is like a video, capturing what happened between two snapshots.

Revenue – Expenses ≡ Income

 © Hannes Wagner

More background

- We do not need to understand all of accounting for a finance application, but a basic understanding is needed.
- If you have never taken an accounting class, you will need more than the material briefly visited here to make sense of what we are covering. You have a huge amount of choice.
- Two suggestions
  - Accounting primer by Aswath Damodaran, nice and brief, available **here** (and see his website generally for useful and free material: http://people.stern.nyu.edu/adamodar).
  - Accounting and financial statements on Khan Academy, available **here**

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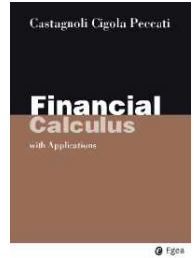
The basics – background reading

- The basis of financial mathematics is provided in Castagnoli, Cigola and Peccati "Financial Calculus with Applications", EGEA 2013, ISBN 9788823821743. Available from EGEA
- The key chapters for this course are
  - 1 Common financial laws
  - 2 Financial laws depending on one variable
  - 3 Financial laws depending on two variables
  - 4 Interest rate term structure
  - 8 Net Present Value
  - 13 Incoherent methods: a short review



A key concept: Intertemporal consumption

- Suppose you receive $100,000 in cash today.
- This is the only income you have to pay for current and future consumption.
- There are only 2 points in time, today (i.e. the current year), and tomorrow (i.e. the next year).
- Suppose further there are no investment projects, banks, or other opportunities to invest your money.
- You may however story money costlessly and without risk for future use.
- What is your intertemporal budget constraint?

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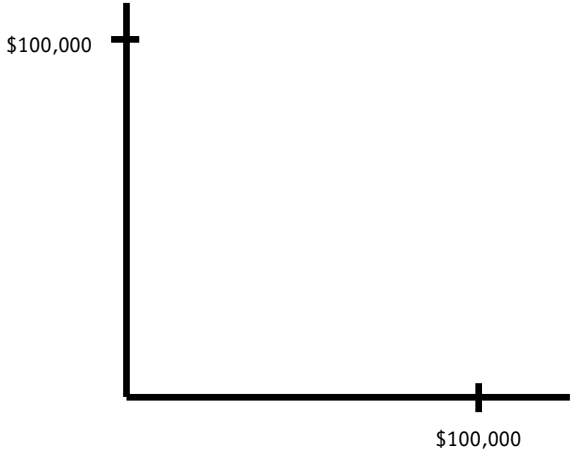
Here we go...



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The time value of money

- Now suppose you receive $100,000 in cash today, again this is your only income. You can invest money today and earn 7% interest a year.
- What is your intertemporal budget constraint now?

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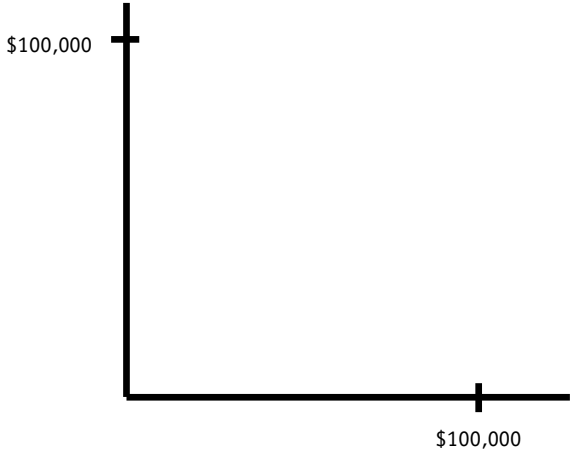
Similar, but different!



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Two ideas follow from this: Present value and future value

- **Future Value:** Amount to which an investment will grow after earning interest
  - Amount today plus interest is amount tomorrow - > Future value
- **Present Value:** Value today of a future cash flow
  - Amount tomorrow net of interest is amount today - > Present value
- Once we have this, we can introduce
  - **Discount Rate:** Interest rate used to compute present values of future cash flows.
  - **Discount Factor:** Present value of a $1 future payment.

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Present Value

- Discounting a cash flow converts it into present value and does several things. First, once cash flows are converted into present value dollars, they can be aggregated and compared. Second, if present values are calculated correctly, one should be indifferent between the future cash flow and the present value of that future cash flow.
- We can write the **present value (PV) of a cash flow** as follows

$$PV = \frac{C_t}{(1+r)^t}$$

where C_t is a cash flow occurring at time t, r is the discount rate

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Example: Valuing an office building

Assume your plan is to construct and then sell an office building to Bocconi. To make it simple, assume you build today and you sell in Year 1. Is this a good project? How do you decide?

Step 1: Forecast cash flows

Cost of building = C_0 = 100,000,000

Sale price in Year 1 = C_1 = 120,000,000

Step 2: Estimate opportunity cost of capital

If equally risky investments in the capital market offer a return of 5%, then

Cost of capital = r = 5%

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Example cont'

Step 3: Discount future cash flows

$$PV = \frac{C_1}{(1+r)} = \frac{120,000,000}{1+.05} = 114,286,000$$

Step 4: Go ahead if PV of payoff exceeds investment

$$\begin{aligned} NPV &= 114,286,000 - 100,000,000 \\ &= 14,286,000 \end{aligned}$$

Here, it obviously pays off to do the investment project. The discounted future cash inflow is larger than the initial investment. Or, a different way of saying the same thing, the project has positive NPV.

Net Present Value Rule

- We can formulate this simple decision rule: Accept investments that have positive net present value. Period.

<u>Example</u>

Suppose we can invest $50 today and receive $60 in one year. Should we accept the project given a 10% expected return?

$$NPV = -50 + \frac{60}{1.10} = \$4.55$$

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Present value properties

- PVs have the **value additive property**. Meaning, if there are multiple cash flows to be evaluated, we can simply **add** them up:

$$PV = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \ldots.$$

- Sometimes there are **shortcuts** that make it very easy to calculate the present value of an asset that pays off in different periods. These tools allow us to cut through the calculations quickly.

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Perpetuity

A perpetuity is a constant cash flow at regular intervals forever.

$$\text{PV of perpetuity} = \frac{C}{r}$$

Example

What is the present value of $1 billion every year, for all eternity, if you estimate the perpetual discount rate to be 10%??

$$PV = \frac{\$1\,\text{bil}}{0.10} = \$10\text{ billion}$$

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Annuity

An annuity is an asset that pays a fixed sum each year for a specified number of years.

$$\text{PV of annuity} = C \times \left[\frac{1}{r} - \frac{1}{r(1+r)^t} \right]$$

Example

You agree to lease a car for 4 years at $3000 per month. You are not required to pay any money up front or at the end of your agreement. If your opportunity cost of capital is 0.5% per month, what is the cost of the lease?

$$\text{Lease Cost} = 3000 \times \left[\frac{1}{.005} - \frac{1}{.005(1+.005)^{48}} \right]$$

$$Cost = \$127,741$$

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Constant growth perpetuity

A growing perpetuity is a cash flow that is expected to grow at a constant rate forever.

$$PV_0 = \frac{C_1}{r - g}$$

g = the annual growth rate of the cash flow

Example

You win, in a lottery, an eternal cash flow that starts next year at $1,000 and grows by 2% every year. The eternal discount rate is 10 percent. What is the present value of the cash flows?

$$PV = \frac{1,000}{0.1 - 0.02} = 12,500$$

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Constant growth annuity

- A constant growth annuity is simply the difference between two constant growth perpetuities.
- Example: A 3-year growing annuity

	Cash flow						
	Year: 1	2	3	4	5	6 . . .	*Present value*
1. Growing perpetuity A	$1	$1 x (1 + g)	$1 x (1 + g)^2	$1 x (1 + g)^3	$1 x (1 + g)^4	$1 x (1 + g)^5 . . .	$\frac{1}{r-g}$
2. Growing perpetuity B				$1 x (1 + g)^3	$1 x (1 + g)^4	$1 x (1 + g)^5 . . .	$\frac{(1+g)^3}{(r-g)(1+r)^3}$
3. Growing 3-year annuity (1 – 2)	$1	$1 x (1 + g)	$1 x (1 + g)^2				$\frac{1}{r-g} - \frac{(1+g)^3}{(r-g)(1+r)^3}$

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Constant growth annuity

Present value of a constant growth annuity:

$$PV = C\left[\frac{1}{r-g} - \frac{(1+g)^t}{(r-g)(1+r)^t}\right]$$

- Keep in mind that this assumes first cash flow at t=1

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Simple and Compound Interest

- A statement that the interest rate on an asset is 10% per annum may sound straightforward and unambiguous. However, its meaning depends on the way interest is computed.
- There are two main ways to calculate interest over multiple periods:
  - Simple interest – interest is earned only on the amount invested (rarely used in practice)
  - Compound interest – interest is earned on the amount invested <u>an on past interest</u> (the most common way interested is calculated in practice)

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Simple and Compound Interest

The table below illustrates the difference between **simple interest** and **compound interest**. It shows the value of a $100 investment earning 10% annually.

	Simple Interest					Compound Interest				
Year	Starting Balance	+	Interest	=	Ending Balance	Starting Balance	+	Ending Interest	=	Balance
1	$100	+	10	=	$110	$100	+	10	=	$110
2	110	+	10	=	120	110	+	11	=	121
3	120	+	10	=	130	121	+	12.1	=	133.1
4	130	+	10	=	140	133.1	+	13.3	=	146.4
10	190	+	10	=	200	236	+	24	=	259
100	1,090	+	10	=	1,100	1,252,783	+	125,278	=	1,378,061
200	2,090	+	10	=	2,100	17,264,116,042	+	1,726,411,604	=	18,990,527,646
230	2,390	+	10	=	2,400	301,248,505,631	+	30,124,850,563	=	331,373,356,194

 © Hannes Wagner

Compounding frequency

- Compound interest does not have to be calculated annually – it can be calculated at any frequency.
- Again, consider an asset that pays 10% per year.
- With annual compounding, $100 at the end of the year grows to

$$\$100 \times 1.1 = \$110$$

at the end of the first year.

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Compounding frequency

- When the 10% per annum interest rate is computed using semi-annual compounding, it means that 5% is earned every 6 months, with the interest being reinvested. This is convention.
- In this case, $100 grows to

$$\$100 \times 1.05 \times 1.05 = \$110.25$$

at the end of the first year.

  - The difference between $110 and $110.25 may seem small. In case it does, consider, say, the $16 billion bridge loan the Italian oil and gas firm Eni received in summer 2012. If interest were 10% p.a., interest payments over a year would be ($16bn (0.1025-0.1) = $40 million higher with semi-annual instead of annual compounding. This starts to look like real money.

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Discrete versus Continuous Compounding

- Generally speaking, consider an amount *A* invested for *n* years at an interest rate of *R* per annum.
- If the rate is compounded once per year the final amount will be

$$A(1+R)^n$$

- If the rate is compounded *m* times per year the final amount will be

$$A\left(1+\frac{R}{m}\right)^{mn}$$

- As m tends to infinity, at the limit we can use continuous compounding. With continuous compounding it can be shown that an amount *A* invested for *n* years at rate *R* grows to

$$Ae^{Rn}$$

where *e* is the mathematical constant 2.71828

 © Hannes Wagner

Discrete versus Continuous Compounding

Example:

Invest $100 using different compounding frequencies. Interest rate is 10%, per annum

Compounding frequency	m	Value of $100 at the end of 1 year
Annually	1	110
Semiannually	2	110.25
Quarterly	4	110.38
Monthly	12	110.47
Weekly	52	110.51
Daily	365	110.52

From the previous results we can derive two conversion formulas:

Convert discrete to continuous compounding

$$R_c = m \ln\left(1 + \frac{R_m}{m}\right)$$

Convert continuous to discrete compounding

$$R_m = m\left(e^{R_c/m} - 1\right)$$

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Effective and nominal interest rates

- The effective interest rate, effective annual interest rate, annual equivalent rate (AER) or simply effective rate is the interest rate on an asset restated from the nominal interest rate as an interest rate with annual compound interest.
- Intuitively, the effective interest rate is calculated as if compounded annually.
- Generally,

$$R_{eff} = \left(1 + \frac{R_m}{m}\right)^m - 1$$

where R_{eff} is the effective interest rate, R_m is the nominal interest rate, and m is the number of compounding periods.

 © Hannes Wagner

Effective and nominal interest rates

- If we consider the previous example again (nominal rate = 10%):

Compounding frequency	m	Value of $100 at the end of 1 year	Effective rate
Annually	1	110	10.00%
Semiannually	2	110.25	10.25%
Quarterly	4	110.38	10.38%
Monthly	12	110.47	10.47%
Weekly	52	110.51	10.51%
Daily	365	110.52	10.52%

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6 Risk-return trade-off

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Bonds

- A **bond** is an instrument of indebtedness of the bond issuer to bond holders. It is a debt security, under which the issuer owes the holders a debt and, depending on the terms of the bond, is obliged to pay them interest (the coupon) and/or to repay the principal at a later date, called maturity.
- Interest is usually payable at fixed intervals (semiannual, annual, sometimes monthly).
- Bonds are the legal form in which debt is issued in the market (as opposed to a bank loan).

Yield to maturity

- **Yield to maturity** (*YTM*) is the [effective annual] interest rate that equates today's value (price) of a financial instrument with the present value of all its future cash flows.
- *YTM* enables us to compare various financial instruments, regardless of the pattern of their cash flows.

Present value and yield to maturity

- Recall the present value formula:

$$PV = \sum_{t=1}^{t=n} \frac{CF_t}{(1+i)^t}.$$

- Here, i is the YTM.
- If you know PV (value or price today) and all of the cash flows CF_t, then i is the yield to maturity.
- Conversely, if you know all of the cash flows CF_t and the yield to maturity i, then you can calculate PV (price or value today).

Yield: Discount bonds

- **Discount bonds (or zero-coupon bonds)** promise only one payment: At maturity the issuer of the bond pays to the holder of the bond the face value.
- The investor (buyer) obtains the bond at a discount to the face value, so as to compensate for the time value of money.

Yield: Discount bonds

- For a discount bond,

$$P = \frac{F}{(1+i)^n},$$

$$\Rightarrow i = \sqrt[n]{\frac{F}{P}} - 1,$$

  where F is the face value and P is the price (present value) today.
- For a one-year bond (most common), this simplifies into

$$i = \frac{F-P}{P}.$$

Yield: Coupon bonds

- **Coupon bonds** promise a stream of payments: The issuer of the bond periodically pays to the holder of the bond a fixed amount (called coupon) and at maturity also repays the face value of the bond (the principal).
- The amount of coupon is determined by the bond's **coupon rate**, which tells you what percentage of the bond's par value is paid as coupons each period.
- Coupon rate:

$$i_{coupon} = \frac{C}{F},$$

where C is the size of payment and F is the face value of the bond.

Yield: Coupon bonds

- For a coupon bond,

$$PV = \sum_{t=1}^{t=n} \frac{C}{(1+i)^t} + \frac{F}{(1+i)^n}.$$

Yield: Coupon bonds

- Assume you have a 10-year coupon bond with a principal (face value) of €1,000 and a 10% coupon rate. What is its price today, if the YTM equals i?

Yield: Coupon bonds

- First, calculate the amount of the coupon payment from the bond's coupon rate:

$$C = F \times i_{coupon} = 1,000 \times 10\% = 100.$$

- It is now easy to calculate the price:

$$P = \frac{\$100}{1+i} + \frac{\$100}{(1+i)^2} + \frac{\$100}{(1+i)^3} + \ldots + \frac{\$100}{(1+i)^{10}} + \frac{\$1000}{(1+i)^{10}}.$$

Relation between price and yield

- Consider *YTM* on a 10-year coupon bond with a principal of €1,000 and a 10% coupon rate.
- The table below gives the YTM on this bond for different possible bond prices:

Price of Bond (€)	*YTM* (%)
1,200	7.13
1,100	8.48
1,000	10.00
900	11.75
800	13.81

Relation between price and yield

- When a bond is trading at par (its price equals its face value), its yield equals its coupon rate.
- Prices and yields are negatively related.
- The yield is greater than the coupon rate when the bond's price is below its par value. The yield is smaller than the coupon rate when the bond's price is above its par value.

Relation between price and yield

- It is straightforward to see why the value of a bond (price) and its yield to maturity (YTM) are negatively related.
- Recall that

$$PV = \sum_{t=1}^{t=n} \frac{CF_t}{(1+i)^t}.$$

- If i goes up, PV has to go down.
- This is an important observation: **bond prices change as interest rates change**.

Duration

- **Duration** is a measure of the bond's sensitivity to interest rates: It tells you how much the price of the bond will change in response to a change in its yield.
- In other words, duration measures the elasticity of the bond's price to the interest rate.
- Here's the formula (we will go through an example, which will make it easier to understand the formula):

$$DUR = \frac{\sum_{t=1}^{t=n} t \frac{CF_t}{(1+i)^t}}{\sum_{t=1}^{t=n} \frac{CF_t}{(1+i)^t}} = \sum_{t=1}^{t=n} t \frac{\frac{CF_t}{(1+i)^t}}{\sum_{t=1}^{t=n} \frac{CF_t}{(1+i)^t}}.$$

Duration

- The importance of duration comes from the following fact.
- For small changes in interest rates:

$$\%\Delta P \equiv \frac{\Delta P}{P} \approx -DUR\frac{\Delta i}{1+i}.$$

- For small changes in interest rates, the percentage change in the bond's price is negatively related to its duration.

Duration

- Reminder:

$$DUR = \frac{\sum_{t=1}^{t=n} t \frac{CF_t}{(1+i)^t}}{\sum_{t=1}^{t=n} \frac{CF_t}{(1+i)^t}} = \sum_{t=1}^{t=n} t \frac{\frac{CF_t}{(1+i)^t}}{\sum_{t=1}^{t=n} \frac{CF_t}{(1+i)^t}} = \sum_{t=1}^{t=n} t \frac{\frac{CF_t}{(1+i)^t}}{PV}.$$

- Intuitively, duration is a weighted average of the timings of cash flows (weighted maturity).
- Each period is weighted by the share of that period's cash flow's present value in the total present value of the bond.

Duration

- Duration of zero-coupon bond is equal to its time until maturity.
- All else equal, when the maturity of a bond lengthens, its duration rises as well.
- All else equal, when interest rates rise, the duration of a coupon bond falls.

Calculating duration

- Let's calculate the duration of a 5-year 10% coupon bond with a face value of €1,000 when its *YTM* is 20%.

Year	Cash Flow (€)	PV of Cash Flow (€)	Weight (%)	Weighted maturity (years)
1	100	83.33	11.89	0.12
2	100	69.44	9.91	0.20
3	100	57.87	8.26	0.25
4	100	48.23	6.88	0.28
5	100	40.19	5.73	0.29
5	1,000	401.88	57.33	2.87
Total		700.94	100.00	3.99

Duration and changes in interest rates

- Recall that:

$$\%\Delta P \equiv \frac{\Delta P}{P} \approx -DUR\frac{\Delta i}{1+i}.$$

- The greater is the duration of a security, the greater is the percentage change in the market value of the security for a given change in interest rates.

Duration and changes in interest rates

- Consider again the bond from our previous example (a €1,000 face value 5-year 10% coupon bond when its interest rate (YTM) is 20%). What will be the percentage change in its price if interest rates rise by 1 percentage point (from 20% to 21%)?

Duration and changes in interest rates

- Answer:

$$\%\Delta P \approx -3.99 \times \frac{.01}{1+0.2} = -0.03325 = -3.325\%.$$

Duration of a portfolio

- An important (and nice) property of duration is additivity.
- The duration of a portfolio of securities is the weighted average of the durations of individual securities, with the weights reflecting the proportion of the portfolio invested in each.

Duration of a portfolio

- Assume that you have a portfolio of two bonds. One has a duration of 3 years, and the other has a duration of 7 years. The market values of the bonds are €600 and €900, respectively. Both bonds have the face value of €1,000. What is the duration of your portfolio?

Duration of a portfolio

- Answer: 5.4 years, since

$$3 \times \frac{600}{600+900} + 7 \times \frac{900}{600+900} = 5.4.$$

- N.B.1: The weights reflect the values (market prices) of the securities in the portfolio, not their face values.
- N.B.2: When interest rates change, the weights of the bonds in your portfolio will also change (why?), and the duration will have to be recalculated.

Outline

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Risk and return

- It is impossible to predict the future value of a financial asset certainty.
- Therefore, some form of expectations/beliefs must be embedded in the valuation of financial assets.
- The degree of uncertainty embedded in the valuation of financial assets is called **risk**.

Risk and return

- Generally, assets that offer high returns are also the assets associated with high risk.
- This is called the **risk-return** tradeoff: seeking higher returns entails accepting higher risk.
- In other words, most of the time people have to choose either a high return or low risk (but not both).

Risk aversion

- Consider the following financial assets:
  - Security A: pays 80 with probability 100%;
  - Security B: pays 40 with probability 50% or 120 with probability 50%;
  - Security C: pays 30 with probability 50% or 130 with probability 50%.
- Which one do you prefer?

Risk aversion

- **Risk-averse** people prefer safe gambles to risky ones and would choose Security A: It offers the same expected return as the other two, but with no risk.
- Between Security B and Security C, a risk-averse person would choose security B.
- Evidence suggests that most people are risk-averse, especially when the stakes are high.

Risk aversion

- All else equal (i.e., for the same level of risk), **rational investors** prefer assets with higher expected returns.
- All else equal (i.e., for the same level of expected return), **risk-averse investors** prefer assets with lower risk.
- Lower standard deviation (for the same expected return) translates into lower risk, since it means less uncertainty about future payoffs.
- Assets that have a high expected return usually have a high standard deviation of those returns (recall the risk-return tradeoff mentioned previously).